MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 2J42H

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Fixed frequency, 9,345 to 9,405 MHz, pulse type, rated peak power output 7 kw, with integral magnet.

ABSOLUTE RATINGS:

										Pressurization			
Parameter:	Ef	tk	epy	ib	Pi	pi	tpc	Du	T(anode)	Output	Input bushing	Alt	VSWR
Unit:	V	sec	kv	a	W	kw	µs	---	°C	psia	psia	ft	---
Maximum:	7.0	---	6.0	5.5	70	35	2.5	0.002	120	45	45	60,000	1.5
Minimum:	---	120	---	3.7	---	---	---	---	---	---	---	---	---
Notes:	5	5	---	---	---	---	---	---	6, 7	8	8	11	---

PHYSICAL CHARACTERISTICS:

See figure 1
Weight: 3-pound (approximate)
Output coupling: UG-40B/U
Cooling: See notes 6 and 7
Magnet isolation: See note 13

TEST CONDITION:

Parameter:	Ef	tk	tpc	rrv	tfv	ib	epy	Du	VSWR
Unit:	V	sec	µs	kv/µs	µs	a	kv	---	---
Tolerance:	---	---	±10%	±10	±0.1	---	±5%	---	---
	6.0 (Note 2)	--- (Note 5)	0.45	75	0.40	4.5	5.27	0.00036	1.1 (Except where noted)

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MIN	MAX	
	Qualification inspection						
4027	Temperature coefficient	-	T(anode) = 40° to 70°C	ΔF/°C	---	0.25	MHz
1026	Temperature range, operating	5		---	---	---	---
4315	Stability	9, 10		MP	---	0.25	%
1031	High-frequency vibration	-	No voltages	---	---	---	---
4003	Pressurizing	-	45 psia	---	---	---	---
1002	Barometric pressure, reduced	11		---	---	---	---
	Quality conformance inspection, part 1	1					
1301	Heater current	2		If	0.43	0.60	A
4218	Fixed-tuned frequency	6, 7	T(anode) = 40° ±10°C	F	9,345	9,405	MHz
4250	Power output	-	t = 300 (max)	Po	2.5	4.0	W
4308	RF bandwidth	3, 4		BW	---	2.0 tpc	MHz
4310	Pulling factor	-		ΔF	---	20	MHz
	Quality conformance inspection, part 2						
1031	Low-frequency vibration	-	No voltages	---	---	---	---
4315	Stability	9, 10		MP	---	0.25	%
	Quality conformance inspection, part 3						
---	Life test	12	Group D (cycled)	t	250	---	hrs
---	Life-test end points:						
4250	Power output	-		Po	2.0	---	W
4308	RF bandwidth	3, 4		BW	---	2.5 tpc	MHz
4315	Stability	9, 10		MP	---	0.5	%

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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.

2. A tk of 180 seconds minimum, with an Ef of 6.3 volts is required before test.

3. The tube current shall be adjusted between 3.7 and 5.5 amperes for widest spectrum.

4. The rf bandwidth shall be within the limits specified, when a VSWR of 1.5/1.0 is introduced in the load at a maximum distance of 0.4 meter from the tube coupling flange, the phase adjusted to produce maximum spectrum degradation.

5. The maximum heater voltage specified is for nonoscillating conditions. The heater surge current shall not exceed 3 amperes. The cathode heating time shall be a minimum of 120 seconds at an ambient temperature greater than 0°C, and a minimum of 180 seconds at temperatures between 0° and -55°C. The heater voltage shall be reduced within 3 seconds after applying pulse voltage according to the following formula:

$$Ef = 6.3\ (1 - 0.03\ Ib) \text{ volts}$$

Where:

Ib = mAdc

6. The anode temperature shall be measured at the point indicated on figure 1.

7. The anode shall be cooled to the temperature specified by a suitable flow of air over the anode body and waveguide attachment brackets which serve as cooling fins.

8. The gas used in pressurization shall provide properties at least equal to that of the clean dry air at the pressure specified herein.

9. This test shall be performed into a 1.5/1 VSWR adjusted in phase to produce maximum instability.

10. Stability shall be measured in terms of the average number of output pulses missing expressed as a percentage of the number of input pulses applied during the period of observation. Pulses shall be considered to be missing if, due to any cause, the rf energy is less than 70 percent of the normal energy level in the frequency range of 9,345 to 9,405 MHz. The percentage of missing pulses (MP) shall not exceed the amount specified herein during the last 3 minutes of a test interval not exceeding 6 minutes. The stability test shall be started 60 seconds after high voltage is applied.

11. The tube shall be operated in a chamber evacuated to simulate pressure conditions of 60,000 feet altitude (54.1 mm Hg absolute). The VSWR shall be 1.3 minimum, varying through all phases. There shall be no evidence of arc-overs during this test.

12. The conditions for cycled life test shall be as follows:

Condition	ib	Ef	Duration
Standby	0	6.3 V	2 minutes
	4.5 amps	6.0 V	1 hour
off	0	0	13 minutes

13. In handling and mounting the magnetron, care shall be exercised to prevent demagnetization. Ferromagnetic materials and energized magnets shall not be brought within 2 inches of the tube.

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NOTES: -Continued

14. Changes from previous issue. The extent of changes (deletions, additions, etc.) precludes the annotation of the individual changes from the previous issue of this document.

Custodians:
Army - EL
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Review activities:
Air Force - 80, 84
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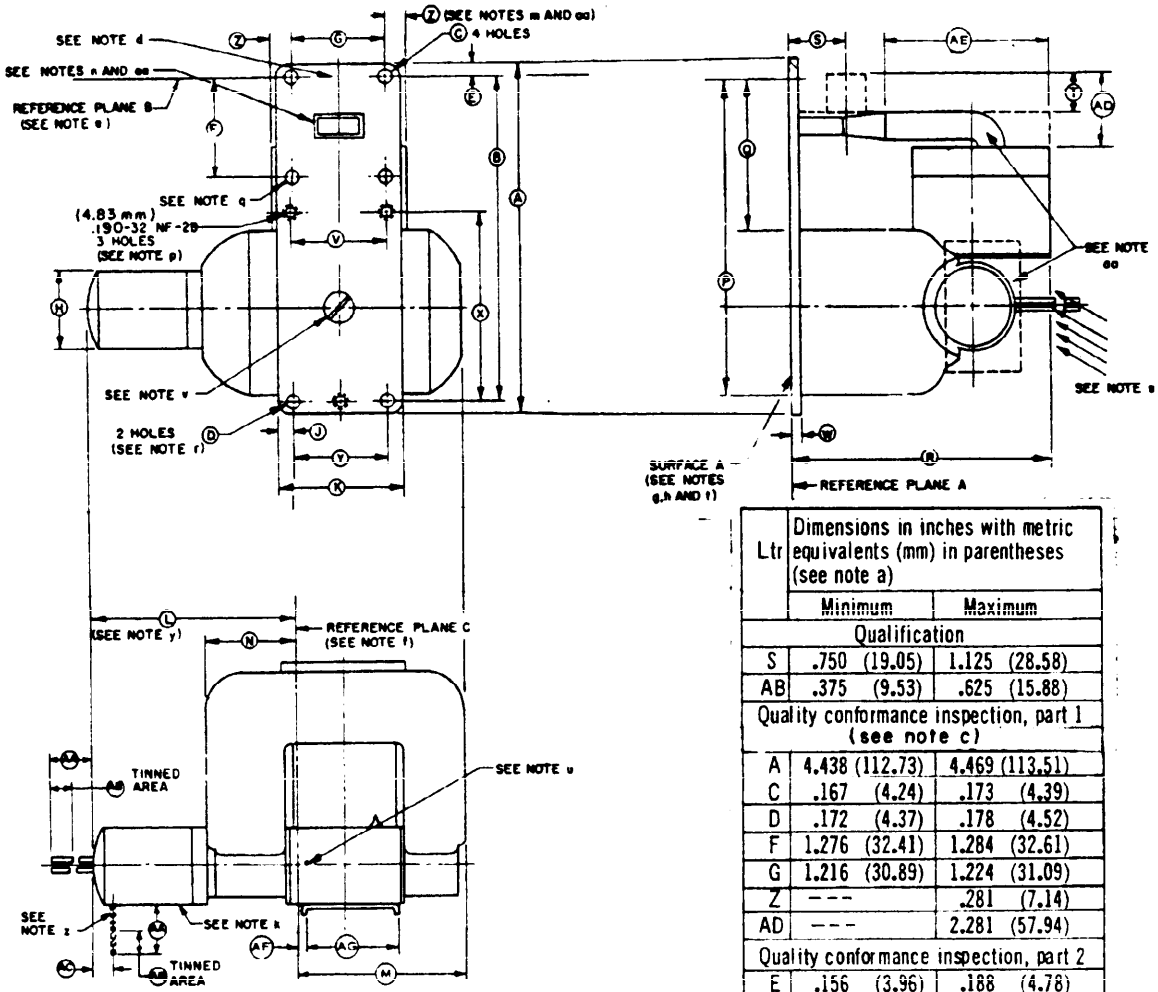
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity: Navy - EC

Agent: DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note a)	
	Minimum	Maximum
Qualification		
S	.750 (19.05)	1.125 (28.58)
AB	.375 (9.53)	.625 (15.88)
Quality conformance inspection, part 1 (see note c)		
A	4.438 (112.73)	4.469 (113.51)
C	.167 (4.24)	.173 (4.39)
D	.172 (4.37)	.178 (4.52)
F	1.276 (32.41)	1.284 (32.61)
G	1.216 (30.89)	1.224 (31.09)
Z	---	.281 (7.14)
AD	---	2.281 (57.94)
Quality conformance inspection, part 2		
E	.156 (3.96)	.188 (4.78)
H	---	1.000 (25.40)
J	.188 (4.78)	.219 (5.56)
K	1.609 (40.87)	1.641 (41.68)
L	2.688 (68.28)	3.188 (80.98)
M	---	2.188 (55.58)
N	---	1.188 (30.18)
P	---	4.000 (101.60)
Q	2.125 (53.98)	---
R	3.125 (79.38)	3.188 (80.98)
T		.500 (12.70)
Nominal dimensions (see note b)		
B	4.103 (104.22)	
V	1.250 (31.75)	
W	.125 (3.18)	
X	2.393 (60.78)	
Y	1.220 (30.99)	
AA	8.000 (203.20)	
AC	.250 (6.35)	
AE	1.875 (47.63)	
AF	.047 (1.19)	
AG	1.125 (28.58)	

FIGURE 1. Outline drawing of electron tube type 2J42H.

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NOTES:

a. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
b. Dimensions without tolerances are for information and are not required for inspection purposes.
c. The AQL for the combined mechanical defectives in quality conformance inspection, part 1, shall be 1 percent.
d. This area is gasketed for pressurizing waveguide output, as with coupler UG-40B/U, in accordance with MIL-F-3922/59, and shall be tested in accordance with method 4003.
e. Reference plane B passes through centers of two top holes of mounting plate as shown and is perpendicular to reference plane A.
f. Reference plane C passes through the upper-left hole on mounting plate as shown and is mutually perpendicular to reference planes A and B.
g. With surface A resting on a flat-surface plate, a feeler gage .020 (.51 mm) thick and .125 (3.18 mm) wide shall not enter more than .125 (3.18 mm) at any point. Note x shall apply.
h. Surface A and interior surfaces of waveguide shall be gold plated 10 MSI or silver plated 30 MSI. Plating is not required over parts made of monel, or equivalent, corrosion resistant material. Note w shall apply.
j. Cathode lead shall be identified by letter 'C'.
k. Lead protector shall be potted in a plastic container to prevent high-voltage breakdown.
m. Measured from centerlines of holes C to outer edges of output flange.
n. Position of waveguide hole not specified since tubes are tested and used with coupler UG-40B/U. See note d.
p. Holes shall be located within .015 (.38 mm) of indicated centers. Note x shall apply.
q. Center of this hole shall lie within .004 (.10 mm) of reference plane C. Note x shall apply.
r. Holes shall lie within .005 (.13 mm) of indicated centers. A cylinder .330 (8.38 mm) diameter, centered in holes shown, shall clear the side of the magnet. Note x shall apply.
s. Recommended direction of airblast.
t. End of lead protector and all metal surfaces except surface A shall be painted with heat-resistant paint. Note w shall apply.
u. Anode temperature shall be measured at this point.
v. Soft solder shall be used. As alternate, tip of screw may be soldered.
w. Shall be inspected on basis of qualification inspection.
x. Shall be inspected on basis of quality conformance inspection, part 1.
y. Dimension L may be increased to 3.188 (80.98 mm) maximum in cases defined by note z.
z. Alternative position of leads.
aa. Configurations of tube, waveguide, output window, and magnet are optional with the provision that specified limits shall not be exceeded.

FIGURE 1. Outline drawing of electron tube type 2J42H - Continued.

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